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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Design and Analysis of Algorithms (course)

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## Course outline

About  
NPTEL ()

How does an  
NPTEL  
online  
course  
work? ()

Week 1 :  
Introduction  
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Week 1 :  
Analysis of  
algorithms ()

Week 1 Quiz  
()

Week 2 :  
Searching  
and sorting  
()

# Week 3 Quiz

The due date for submitting this assignment has passed.

**Due on 2025-08-13, 23:59 IST.**

**As per our records you have not submitted this assignment.**

All questions carry equal weightage. You may submit as many times as you like within the deadline. Your final submission will be graded.

1) A connected undirected graph  $G$  has 2775 edges. What can we say about  $n$ , the number of vertices in  $G$ ? **1 point**

☐  $76 \leq n \leq 2775$

☐  $75 \leq n \leq 2775$

☐  $76 \leq n \leq 2776$

☐  $75 \leq n \leq 2776$

No, the answer is incorrect.

Score: 0

Feedback:

*A tree with 2776 vertices has 2775 edges. A complete graph with 75 vertices has 2775 edges.*

Accepted Answers:

$75 \leq n \leq 2776$

2) An airline serves 1000 cities and runs 5500 direct flights each day between these cities. Which of the following is a good data structure to represent the collection of flights? **1 point**

☐ A  $1000 \times 1000$  array  $A$ , where  $A[i][j] = 1$  if there is a direct flight from city  $i$  to city  $j$  and 0 otherwise.

☐ A stack containing values  $(i, j)$  for each pair of cities  $i, j$  for which there is a direct flight from city  $i$  to city  $j$ .

☐ A queue containing values  $(i, j)$  for each pair of cities  $i, j$  for which there is a direct flight from city  $i$  to city  $j$ .

## Week 2 Quiz ( )

## Week 2 Programmin g Assignment ( )

## Week 3 : Graphs ( )

## Week 3 Quiz ( )

## Quiz: Week 3 Quiz (assessment? name=237)

## Week 3 Programmin g Assignment ( )

## Week 4 : Weighted graphs ( )

## Week 4 Quiz ( )

## Week 4 Programmin g Assignment ( )

## Week 5: Data Structures: Union-Find and Heaps ( )

## Week 5 : Divide and Conquer ( )

## Week 5 Quiz ( )

- ☐ A list containing values  $(i, j)$  for each pair of cities  $i, j$  for which there is a direct flight from city  $i$  to city  $j$ .

No, the answer is incorrect.

Score: 0

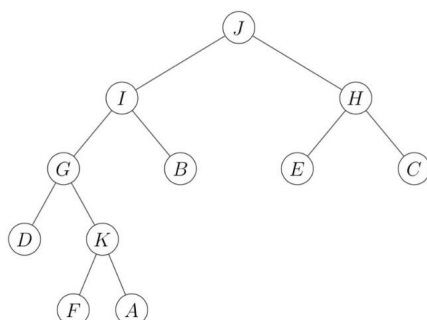
Feedback:

The array will have 1,000,000 entries, while a list will have 11,000 elements. Even allowing for an overhead for pointers etc, the list will be more space-efficient. There is no benefit to using a stack or queue.

Accepted Answers:

A list containing values  $(i, j)$  for each pair of cities  $i, j$  for which there is a direct flight from city  $i$  to city  $j$ .

- 3) Suppose we obtain the following BFS tree rooted at node J for an undirected graph **1 point** with vertices  $\{A, B, C, D, E, F, G, H, I, J, K\}$ .



Which of the following *cannot* be an edge in the original graph?

- ☐ (C, G)  
☐ (B, K)  
☐ (B, F)  
☐ (A, D)

No, the answer is incorrect.

Score: 0

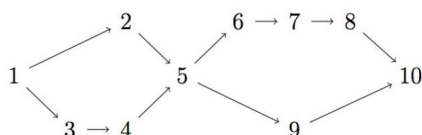
Feedback:

In an undirected graph, all nontree edges must be at the same level or between adjacent levels. (B, F) is an edge across two levels.

Accepted Answers:

(B, F)

- 4) We are interested in topological orderings of the following DAG which satisfy the constraint that whenever 9 appears after 8, 2 must appear after 4. How many such orderings are there? **1 point**



- ☐ 12  
☐ 10  
☐ 9  
☐ 8

**Week 6: Data Structures: Search Trees**  
( )

**Week 6: Greedy Algorithms**  
( )

**Week 6 Quiz**  
( )

**Week 6 Programming Assignment**  
( )

**Week 7: Dynamic Programming**  
( )

**Week 7 Quiz**  
( )

**Week 7 Programming Assignment**  
( )

**Download Videos**  
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**Lecture Material**  
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**Problem Solving Session - July 2025**  
( )

No, the answer is incorrect.

Score: 0

Feedback:

*There are two blocks in parallel,  $\{2\} \parallel \{3,4\}$  and  $\{6,7,8\} \parallel \{9\}$  that can be reordered. There are 3 ways of ordering the right block, of which only one, 6-7-8-9 has 9 after 8. There are 3 ways ordering the left block, of which only one, 3-4-2, has 2 after 4. So there are  $3 \times 3 = 9$  unconstrained topological orders, plus one that meets the constraint.*

Accepted Answers:

10

5) Applying for permits to put up a factory is an 11 step process. Some steps depend on others, as described below. **1 point**

- Step 1 must be completed before steps 3 and 4 start.
- Step 2 must be completed before steps 3, 6 and 7 start.
- Step 3 must be completed before step 7 starts.
- Step 4 must be completed before step 5 starts.
- Step 5 must be completed before step 7 starts.
- Step 7 must be completed before steps 8 and 9 start.
- Step 9 must be completed before steps 10 and 11 start.

Each step takes a week to complete. What is the minimum number of weeks required to get all the permits in place?

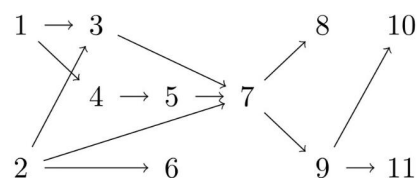
- ☐ 4
- ☐ 6
- ☐ 8
- ☐ 9

No, the answer is incorrect.

Score: 0

Feedback:

*The constraints describe the following dag.*



*A typical longest path is  $1 \rightarrow 4 \rightarrow 5 \rightarrow 7 \rightarrow 9 \rightarrow 10$  which requires 6 weeks.*

Accepted Answers:

6